The transitive suffix in the later-cut equality case

Human Compression project: L044 continuation

Lemma 1 (An anchored action on pairs)**.** *Let v be a fixed-point-free involution on a set A of size $2t$, $t \geq 2$. Let $Q \leq \mathrm{Alt}(A)$ be transitive, centralize v, and have an involutory string marking $q_1, \ldots, q_{t-1}$. Suppose every maximal omission of the induced marking on the t pairs of v is intransitive. Suppose a point $a \in A$ is fixed by $q_2, \ldots, q_{t-1}$. Then the pair image is the natural S_t with its successive transposition marking, anchored at the pair containing a. Its kernel is nontrivial. If t is even, $v \in Q$. If t is odd, Q is the entire group of even permutations centralizing v, namely $C_2^{t-1} \rtimes S_t$ in this literal action on A.*

Proof. Choose, for each pair-image generator, an edge crossing the orbits of its omission. One edge per label forms a forest: a cycle would contain an avoiding path between the endpoints of one selected edge. There are $t-1$ edges on t vertices, so this forest is a tree T.

Removing its edge of label i gives exactly the two pair-orbits of that omission. Indeed all other tree edges belong to the omission, so connect each of the two tree components, and intransitivity prevents their joining. For a fixed label i, its image preserves both sides of every cut obtained by deleting an edge with a different label. Intersecting those cuts leaves only one nonsingleton class: the two ends of edge i. Thus the image is precisely that transposition.

Nonadjacent labels commute, so their tree edges are disjoint. A vertex cannot have three incident edges, since three distinct indices cannot be pairwise consecutive. Hence T is a path. The labels along it are successive indices, in one of the two orders. The anchor pair is fixed by all labels except 1, so it is the endpoint incident with label 1. The image of $\langle q_2, \ldots, q_{t-1} \rangle$ is consequently its full stabilizer S_{t-1} in S_t.

Write N for the kernel on pairs. If $N = 1$, the subgroup $\langle q_2, \ldots, q_{t-1} \rangle$ is the full pair stabilizer in Q. It fixes a, so the orbit of a has size at most t, contrary to transitivity on $2t$ points. Therefore $N \neq 1$.

The centralizer of v is the permutation group on its pairs with independent pair flips, $C_2^t \rtimes S_t$. A permutation of the pairs is even on the underlying $2t$ points. Hence N is a nonzero S_t-invariant subspace of

$$E = \{(x_1, \ldots, x_t) \in \mathbb{F}_2^t : x_1 + \cdots + x_t = 0\}.$$

The invariance follows from the full pair image; conjugation of flips permutes their coordinates. If N contains a vector whose support is nonempty and not all coordinates, exchange one coordinate in its support with one outside. Adding the vector to its image gives a vector supported on precisely those two coordinates. Conjugating gives every such vector, which together span E. Thus either $N = E$, or its only nonzero vector is the all-ones vector. For even t, both possibilities contain the all-ones flip v. For odd t, the all-ones vector is not in E, so $N = E$. The order of Q is then $2^{t-1}t!$, equal to that of the full even centralizer containing it; hence they coincide. □

Theorem 2 (Exclusion of the transitive equality branch)**.** *Retain the complete critical-head hypotheses, actual IP and all intransitive ambient omissions. Suppose a later unique interior cut has adjacent first shared label g and crossing label $j = g + 1$. Write its two omission orbits as $C = C_g$ and A. Suppose the exact reversed support equality gives*

$$|A| = 2t, \quad t = r - g - 1, \quad \mathrm{Fix}(P_{g-1}) = A \cup \{b\}, \qquad P_k = G_{\leq k},$$

where $b \in C$ is the crossing endpoint. Then $Q = G_{>j}$ cannot be transitive on A.

Proof. Suppose Q is transitive on A. The actual unique-cut lemma makes it fix C pointwise. Its rank is $r-1-j=t-1$, and its action on A is faithful and even. The shared generator ρ_g acts nontrivially on A and centralizes Q. A centralizing permutation of a transitive action either has no fixed point or is the identity; hence v is a fixed-point-free involution. The unique-crossing anchor on the A side is fixed by all Q generators except its first one, by the complete acquired crossing lemma.

Every pair-image omission in Q is intransitive. For if omitting a mark q left its pair action transitive, adjoining v would make the action on A transitive. In the full ambient omission G_q, that joining action is supplied by ρ_g; it preserves A and restricts there to v. Thus A lies in a single G_q-orbit. The omitted mark q is supported wholly on A, so adjoining it cannot join any G_q-orbits. This contradicts full transitivity and the intransitive omission hypothesis. All hypotheses of Lemma 1 now hold.

If t were even, the lemma would give $v \in Q$. On the other hand the exterior image of P_g on A is exactly $\langle v \rangle$, since P_{g-1} fixes A pointwise. Its head orbit is C, and the acquired exact parity-fibre formula applies. The involution v is even when t is even, so that formula places its pure permutation on A in P_g as well. Actual IP for the disjoint index sets $\{0,\dots,g\}$ and $\{g+2,\dots,r-1\}$ gives $P_g \cap Q = 1$, a contradiction. Therefore t is odd, necessarily $t \geq 3$, and the lemma makes Q the full even centralizer of v.

Let $\{a, a'\}$ be the anchor pair, with $a' = a^v$, and let Q_1 omit the first distinguished generator of Q. Its pair image is the full S_{t-1} on the other pairs, while every one of its generators fixes a, and hence a' by centralization. Since $t-1 \geq 2$, it has no fixed point in any other pair. Thus its complete pointwise fixed set is

$$\mathrm{Fix}(Q_1) = C \cup \{a, a'\}.$$

The crossing $s = \rho_{g+1}$ centralizes Q_1, so preserves this fixed set. It exchanges a with b. It must fix a': it cannot send a' into C, which would be a second crossing, and cannot send it to a, whose image is already b. Consequently s also preserves $A \setminus \{a, a'\}$.

The old prefix P_g preserves that complement too, because its exterior image is $\langle v \rangle$. Since P_g is transitive on C, the crossing attaches a to its head orbit and ρ_g then attaches a'. No further point can join. Hence the actual next head orbit is exactly

$$C_{g+1} = C \cup \{a, a'\}.$$

The exterior image of P_{g+1} contains the restriction $v' = v|_{A\setminus\{a,a'\}}$, coming from ρ_g. This is a nonidentity even involution: it flips $t-1 \geq 2$ pairs. The exact parity-fibre formula therefore puts its pure permutation in P_{g+1}. But the entire even-flip kernel belongs to Q, so $v' \in Q$ also. This contradicts actual IP for the disjoint intervals $\{0,\dots,g+1\}$ and $\{g+2,\dots,r-1\}$. The odd case is excluded as well. □

Attribution and scope. The orbit forest and cut saturation reuse the acquired actual-orbit method, here with one edge per pair-image label. The elementary invariant-subspace argument and the two successive pure-fibre/IP exclusions are the present continuation. The complete prefix and unique-crossing proofs remain paid. No classification of projected C-groups is assumed. This theorem applies to the equality interfaces from the separately submitted first-later-cut reduction, but states its hypotheses explicitly and does not depend on that packet's acceptance. The transitive equality suffix is excluded in full. The intransitive suffix, endpoint cuts, opposite heads and terminal singleton still require analysis. No whole-original, question, formal or numerical credit is claimed.